

Homework

Sec. 8.2 2,4(g), 20, 30, 35

Divide-and-Conquer Algorithms and Recurrence Relations

Section 8.3

Section Summary

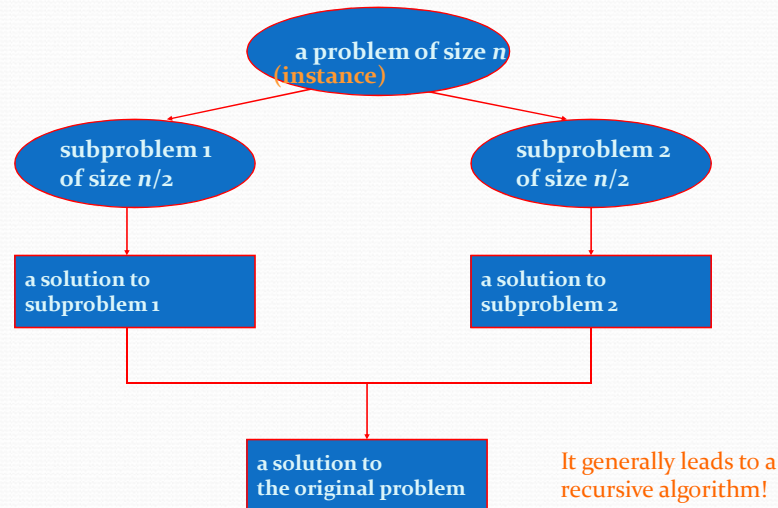
- Divide-and-Conquer Algorithms and Recurrence Relations
- Examples
 - Binary Search
 - Merge Sort
 - Fast Multiplication of Integers
- Master Theorem
- Closest Pair of Points (*not covered yet in these slides*)

Divide-and-Conquer

The most-well known algorithm design strategy:

1. **Divide** instance of problem into two or more smaller instances
2. Solve smaller instances recursively
3. **Conquer** the solution to original (larger) instance by combining these solutions

Divide-and-Conquer (cont.)



Divide-and-Conquer Algorithmic Paradigm

Definition: A *divide-and-conquer algorithm* works by first *dividing* a problem into one or more instances of the same problem of smaller size and then *conquering* the problem using the solutions of the smaller problems to find a solution of the original problem.

Examples:

- Binary search, covered in Chapters 3 and 5: It works by comparing the element to be located to the middle element. The original list is then split into two lists and the search continues recursively in the appropriate sublist.
- Merge sort, covered in Chapter 5: A list is split into two approximately equal sized sublists, each recursively sorted by merge sort. Sorting is done by successively merging pairs of lists.

Divide-and-Conquer Recurrence Relations

- Suppose that a recursive algorithm divides a problem of size n into a (a is a positive integer) subproblems.
- Assume each subproblem is of size n/b . (b is a positive integer)
- Suppose $g(n)$ extra operations are needed in the conquer step.
- Then $f(n)$ represents the number of operations to solve a problem of size n satisfies the following recurrence relation:

$$f(n) = af(n/b) + g(n)$$

- This is called a *divide-and-conquer recurrence relation*.

Example: Binary Search

- Binary search reduces the search for an element in a sequence of size n to the search in a sequence of size $n/2$. Two comparisons are needed to implement this reduction;
 - one to decide whether to search the upper or lower half of the sequence and
 - the other to determine if the sequence has elements.
- Hence, if $f(n)$ is the number of comparisons required to search for an element in a sequence of size n , then

$$f(n) = f(n/2) + 2$$

when n is even.

Example: Merge Sort

- The merge sort algorithm splits a list of n (assuming n is even) items to be sorted into two lists with $n/2$ items. It uses fewer than n comparisons to merge the two sorted lists.
- Hence, the number of comparisons required to sort a sequence of size n , is no more than $M(n)$ where

$$M(n) = 2M(n/2) + n.$$

Example: Fast Multiplication of Integers

- An algorithm for the fast multiplication of two $2n$ -bit integers (assuming n is even) first splits each of the $2n$ -bit integers into two blocks, each of n bits.
- Suppose that a and b are integers with binary expansions of length $2n$. Let $a = (a_{2n-1}a_{2n-2} \dots a_1a_0)_2$ and $b = (b_{2n-1}b_{2n-2} \dots b_1b_0)_2$.
- Let $a = 2^n A_1 + A_0$, $b = 2^n B_1 + B_0$, where $A_1 = (a_{2n-1} \dots a_{n+1})_2$, $A_0 = (a_n \dots a_1a_0)_2$, $B_1 = (b_{2n-1} \dots b_{n+1})_2$, $B_0 = (b_n \dots b_1b_0)_2$.
- The algorithm is based on the fact that ab can be rewritten as:
 - $ab = 2^{2n} A_1 B_1 + 2^n (A_1 B_0 + A_0 B_1) + A_0 B_0$.
 - $ab = (2^{2n} + 2^n) A_1 B_1 + 2^n (A_1 - A_0)(B_0 - B_1) + (2^n + 1) A_0 B_0$.
- This identity shows that the multiplication of two $2n$ -bit integers can be carried out using three multiplications of n -bit integers, together with additions, subtractions, and shifts.
- Hence, if $f(n)$ is the total number of operations needed to multiply two n -bit integers, then

$$f(2n) = 3f(n) + Cn$$

where Cn represents the total number of bit operations; the additions, subtractions and shifts that are a constant multiple of n -bit operations.

Estimating the Size of Divide-and-Conquer Functions

Theorem 1: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + c$$

whenever n is divisible by b , where $a \geq 1$, b is an integer greater than 1, and c is a positive real number.

Then

$$f(n) \text{ is } \begin{cases} O(n^{\log_b a}) & \text{if } a > 1 \\ O(\log n) & \text{if } a = 1. \end{cases}$$

Furthermore, when $n = b^k$ and $a \neq 1$, where k is a positive integer,

$$f(n) = C_1 n^{\log_b a} + C_2$$

where $C_1 = f(1) + c/(a-1)$ and $C_2 = -c/(a-1)$.

$$f(n) = af(n/b) + c$$

Theorem 1:

If $a = 1$, $f(n) \in \Theta(\log n)$

If $a > 1$, $f(n) \in \Theta(n^{\log_b a})$

$$f(n) = af(n/b) + C = a^2 f(n/b^2) + aC + C$$

$$= a^3 f(n/b^3) + a^2 C + aC + C$$

=.....

$$= a^k f(n/b^k) + a^{k-1} C + a^{k-2} C + \dots + C$$

Let $n = b^k$, $n/b^k = 1$,

$$= a^k f(1) + \sum_{j=0}^{k-1} a^j c$$

$$f(n) = af(n/b) + C$$

If $a = 1$, then $f(n) = f(1) + Ck$

Since $n = b^k$, $k = \log_b n$, hence

$$f(n) = f(1) + c \log_b n$$

When $a > 1$ and $n = b^k$,

$$f(n) = a^k f(1) + \frac{c(a^k - 1)}{a - 1}$$

$$= a^k [f(1) + c/(a - 1)] - c/(a - 1)$$

$$= C_1 n^{\log_b a} + C_2$$

$$a^k = a^{\log_b n} = n^{\log_b a}$$

Complexity of Binary Search

Binary Search Example: Give a big-O estimate for the number of comparisons used by a binary search.

Solution: Since the number of comparisons used by binary search is $f(n) = f(n/2) + 2$ where n is even, by Theorem 1, it follows that $f(n)$ is $O(\log n)$.

Estimating the Size of Divide-and-conquer Functions (*continued*)

Theorem 2. Master Theorem: Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + cn^d$$

whenever $n = b^k$, where $a \geq 1$, b is an integer greater than 1, k is a positive integer greater than 1, and c and d are real numbers with c positive and d nonnegative. Then

$$f(n) \text{ is } \begin{cases} O(n^d) & \text{if } a < b^d, \\ O(n^d \log n) & \text{if } a = b^d, \\ O(n^{\log_b a}) & \text{if } a > b^d. \end{cases}$$

Complexity of Merge Sort

Merge Sort Example: Give a big-O estimate for the number of comparisons used by merge sort.

Solution: Since the number of comparisons used by merge sort to sort a list of n elements is less than $M(n)$ where $M(n) = 2M(n/2) + n$, by the master theorem $M(n)$ is $O(n \log n)$.

Complexity of Fast Integer Multiplication Algorithm

Integer Multiplication Example: Give a big- O estimate for the number of bit operations used needed to multiply two n -bit integers using the fast multiplication algorithm.

Solution: We have shown that $f(n) = 3f(n/2) + Cn$, when n is even, where $f(n)$ is the number of bit operations needed to multiply two n -bit integers. Hence by the master theorem with $a = 3$, $b = 2$, $c = C$, and $d = 1$ (so that we have the case where $a > b^d$), it follows that $f(n)$ is $O(n^{\log 3})$.

Note that $\log 3 \approx 1.6$. Therefore the fast multiplication algorithm is a substantial improvement over the conventional algorithm that uses $O(n^2)$ bit operations.

Homework

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Generating Functions

Section 8.4

Section Summary

- Generating Functions
- Useful Generating Functions
- Counting Problems and Generating Functions
- Solving Recurrence Relations Using Generating Functions
- Proving Identities Using Generating Functions

Generating Functions

Definition: The *generating function* for the sequence $a_0, a_1, \dots, a_k, \dots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k.$$

A generating function is a clothesline on which we hang up a sequence of numbers for display.

— Herbert Wilf, *Generating functionology* (1994)

- ◆ Questions about the convergence of these series are ignored.
- ◆ The fact that a function has a unique power series around $x = 0$ will also be important

Generating Functions

Examples:

- The sequence $\{a_k\}$ with $a_k = 3$ has the generating function $\sum_{k=0}^{\infty} 3x^k = \frac{3}{1-x}$ for $|x| < 1$
- The sequence $\{a_k\}$ with $a_k = k + 1$ has the generating function $\sum_{k=0}^{\infty} (k+1)x^k$.
- The sequence $\{a_k\}$ with $a_k = 2^k$ has the generating function $\sum_{k=0}^{\infty} 2^k x^k$.

Generating Functions for Finite Sequences

- Generating functions for finite sequences of real numbers can be defined by extending a finite sequence $a_0, a_1, \dots, a_n$ into an infinite sequence by setting $a_{n+1} = 0, a_{n+2} = 0$, and so on.
- The generating function $G(x)$ of this infinite sequence $\{a_n\}$ is a polynomial of degree n because no terms of the form $a_j x^j$ with $j > n$ occur, that is,

$$G(x) = a_0 + a_1 x + \cdots + a_n x^n.$$

Generating Functions for Finite Sequences (continued)

Example: What is the generating function for the sequence 1,1,1,1,1,1?

Solution: The generating function of 1,1,1,1,1,1 is

$$1 + x + x^2 + x^3 + x^4 + x^5.$$

By Theorem 1 of Section 2.4, we have

$$(x^6 - 1)/(x - 1) = 1 + x + x^2 + x^3 + x^4 + x^5$$

when $x \neq 1$.

Consequently $G(x) = (x^6 - 1)/(x - 1)$ is the generating function of the sequence.

Generating Functions for Finite Sequences (continued)

Example:

Let $a_k = C(m, k)$, $k = 0, 1, 2, \dots, m$. The generating function for this sequence is

$$G(x) = C(m, 0) + C(m, 1)x + C(m, 2)x^2 + \dots + C(m, m)x^m = (1 + x)^m$$

Counting Problems and Generating Functions

Generating functions can be used to solve a wide variety of counting problems, such as

- ✓ Count the number of combinations from a set when repetition is allowed and additional constraints exist.
- ✓ Count the number of permutations

Counting Problems and Generating Functions

Example: Find the number of solutions of

$$e_1 + e_2 + e_3 = 17,$$

where e_1, e_2 , and e_3 are nonnegative integers with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$.

Solution: The number of solutions is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5) (x^3 + x^4 + x^5 + x^6) (x^4 + x^5 + x^6 + x^7).$$

This follows because a term equal to x^{17} is obtained in the product by picking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3} , where $e_1 + e_2 + e_3 = 17$.

There are three solutions since the coefficient of x^{17} in the product is 3.

[Example] Determine the number of ways to insert tokens worth \$1, \$2 and \$5 into a vending machine to pay for an item that costs r dollars in both the cases when the order in which the tokens are inserted does not matter and when the order does matter.

Solution:

(1) The order in which the tokens are inserted does not matter

$$G(x) = (1 + x + x^2 + x^3 + \cdots)(1 + x^2 + x^4 + x^6 + \cdots)(1 + x^5 + x^{10} + x^{15} + \cdots)$$

The coefficient of x^r in the expansion of $G(x)$ is the solution of this problem.

(2) The order in which the tokens are inserted does matter

- ❖ The number of ways to insert exactly n tokens to produce a total of r \$ is the coefficient of x^r in $(x + x^2 + x^5)^n$
- ❖ Since any number of tokens may be inserted, the number of ways to produce r \$ using \$1,\$2 and \$5 tokens, is the coefficient of x^r in

$$1 + (x + x^2 + x^5) + (x + x^2 + x^5)^2 + \cdots = \frac{1}{1 - (x + x^2 + x^5)}$$

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Counting Problems and Generating Functions (*continued*)

Example: Use generating functions to find the number of k -combinations of a set with n elements, i.e., $C(n, k)$.

Solution: Each of the n elements in the set contributes the term $(1 + x)$ to the generating function

$$f(x) = \sum_{k=0}^n a^k x^k.$$

Hence $f(x) = (1 + x)^n$ where $f(x)$ is the generating function for $\{a^k\}$, where a^k represents the number of k -combinations of a set with n elements.

By the binomial theorem, we have

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k,$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Hence,

$$C(n, k) = \frac{n!}{k!(n-k)!}.$$

Useful Facts About Power Series

【Theorem 1】 Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $g(x) = \sum_{k=0}^{\infty} b_k x^k$. Then

$$(1) f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k$$

$$(2) \alpha \cdot f(x) = \sum_{k=0}^{\infty} \alpha \cdot a_k x^k \quad \alpha \in R$$

$$(3) x \cdot f'(x) = \sum_{k=0}^{\infty} k \cdot a_k x^k$$

$$(4) f(\alpha x) = \sum_{k=0}^{\infty} \alpha^k \cdot a_k x^k$$

$$(5) f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

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Useful Facts About Power Series

【Theorem 1】 Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$. *Proof:*

$$(1) f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k$$

$$(2) \alpha \cdot f(x) = \sum_{k=0}^{\infty} \alpha \cdot a_k x^k \quad \alpha$$

$$(3) x \cdot f'(x) = \sum_{k=0}^{\infty} k \cdot a_k x^k$$

$$(4) f(\alpha x) = \sum_{k=0}^{\infty} \alpha^k \cdot a_k x^k$$

$$(5) f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

$$\sum_{k=0}^{\infty} k a_k x^k = \sum_{k=0}^{\infty} a_k \cdot x \cdot k x^{k-1}$$

$$= x \sum_{k=0}^{\infty} a_k (x^k)'$$

$$= x \left(\sum_{k=0}^{\infty} a_k x^k \right)'$$

$$= x f'(x)$$

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Useful Facts About Power Series

Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $g(x) = \sum_{k=0}^{\infty} b_k x^k$. **Then**

$$(1) \quad f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k$$

$$(2) \quad \alpha \cdot f(x) = \sum_{k=0}^{\infty} \alpha \cdot a_k x^k \quad \alpha \in R$$

$$(3) \quad x \cdot f'(x) = \sum_{k=0}^{\infty} k \cdot a_k x^k$$

$$(4) \quad f(\alpha x) = \sum_{k=0}^{\infty} \alpha^k \cdot a_k x^k$$

$$(5) \quad f(x)g(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

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Useful Facts About Power Series

Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$

$$(1) \quad f(x) + g(x) =$$

$$(2) \quad \alpha \cdot f(x) = \sum_{k=0}^{\infty}$$

$$(3) \quad x \cdot f'(x) =$$

$$(4) \quad f(\alpha x) = \sum_{k=0}^{\infty}$$

$$(5) \quad f(x)g(x) =$$

Proof:

$$\sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

$$= a_0 b_0 + (a_0 b_1 + a_1 b_0) x + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^2$$

$$+ \dots + \left(\sum_{j=1}^k a_j b_{k-j} \right) x^k + \dots$$

$$= (a_0 + a_1 x + a_2 x^2 + \dots)(b_0 + b_1 x + b_2 x^2 + \dots)$$

$$= f(x) \cdot g(x)$$

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8.4 Generating Functions

Using the above properties, the generating functions of some sequences can be obtained easily.

【Example】 What is the generating function for the sequence 0,1,2,3,4,5,...?

Solution:

$$b_k = k$$

$$\begin{aligned} G(x) &= \sum_{k=0}^{\infty} kx^k \\ &= x\left(\frac{1}{1-x}\right)' \\ &= \frac{x}{(1-x)^2} \end{aligned}$$

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【Example】 Suppose that the generating function of the sequence: $a_0, a_1, a_2, \dots, a_n, \dots$ is $G(x)$. What is the generating function for the sequence $b_k = \sum_{i=0}^k a_i$?

Solution:

$$a_k \leftrightarrow G(x), \quad b_k \leftrightarrow$$

$$c_k = 1$$

$$b_k = \sum_{i=0}^k a_i$$

$$= \sum_{i=0}^k a_i \times c_{k-i}$$

$$\underline{F(x) = G(x) \cdot \frac{1}{1-x}}$$

For example:

$$1, 1, 1, \dots \longleftrightarrow \frac{1}{1-x}$$



$$1, 2, 3, 4, \dots, k+1, \dots \longleftrightarrow \frac{1}{(1-x)^2}$$

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【Example】 Let $f(x) = \frac{1}{1-4x^2}$. Find the coefficient $a_0, a_1, a_2, \dots, a_n, \dots$ in the expansion $f(x) = \sum_{k=0}^{\infty} a_k x^k$.

Solution:

$$f(x) = \frac{1}{1-4x^2} = \frac{1}{(1-2x)(1+2x)} = \frac{1}{2} \left(\frac{1}{1-2x} + \frac{1}{1+2x} \right)$$

$\updownarrow$
 $\updownarrow$
 $\updownarrow$

$a_k = \frac{1}{2} (2^k + (-2)^k) = \begin{cases} 2^k & k \text{ is even} \\ 0 & k \text{ is odd} \end{cases}$

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✧ The extended binomial coefficient

Recall:

$$\binom{m}{k} = C(m, k) = \frac{m!}{k!(m-k)!}$$

Where m, k are nonnegative integers, $k \leq m$

【Definition 2】 Let u be a real number and k a nonnegative integer. Then the *extended binomial coefficient* is defined by

$$\binom{u}{k} = \begin{cases} u(u-1)\cdots(u-k+1)/k! & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$$

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【Example】 (1) $\binom{1/2}{3} = ?$ (2) $\binom{-n}{k} = ?$

Solution:

$$(1) \binom{1/2}{3} = \frac{(1/2)(1/2-1)(1/2-2)}{3!} = 1/16$$

$$\begin{aligned} (2) \binom{-n}{k} &= \frac{(-n)(-n-1)\dots(-n-k+1)}{k!} \\ &= \frac{(-1)^k n(n+1)\dots(n+k-1)}{k!} \\ &= (-1)^k C(n+k-1, k) \end{aligned}$$

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✧ The extended Binomial Theorem

【Theorem 2】 Let x be a real number with $|x| < 1$ and let u be a real number. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

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【Example】 Find the generating functions for

$$(1+x)^{-n} \text{ and } (1-x)^{-n}$$

where n is a positive integer, using the extended Binomial Theorem.

Solution:

By the extended Binomial Theorem, it follows that

$$\begin{aligned} (1+x)^{-n} &= \sum_{k=0}^{\infty} \binom{-n}{k} x^k \\ &= \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) x^k \\ (1-x)^{-n} &= \sum_{k=0}^{\infty} \binom{-n}{k} (-x)^k \\ &= \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) (-1)^k x^k \\ &= \sum_{k=0}^{\infty} C(n+k-1, k) x^k \end{aligned}$$

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Useful Generating Functions

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \dots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \dots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \dots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \dots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \dots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \dots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \dots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \dots$	$k+1$
$\frac{1}{(1-x)^3} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \dots$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\frac{1}{(1+x)^3} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \dots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^3} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2x^2 + \dots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$(-1)^{k+1}/k$

Note: The series for the last two generating functions can be found in most calculus books when power series are discussed.

Sequence	Generating function
(1) $C(n, k)$	$\sum_{k=0}^{\infty} C(n, k)x^k = (1+x)^n$
(2) $C(n, k)a^k$	$(1+ax)^n$
(3) $1, 1, \dots, 1$	$1+x+x^2+\dots+x^n = \frac{1-x^{n+1}}{1-x}$
(4) $1, 1, 1, \dots$	$\frac{1}{1-x}$
(5) a^k	$\frac{1}{1-ax}$
(6) $k+1$	$\frac{1}{(1-x)^2}$

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Sequence	Generating function
(7) $C(n+k-1, k)$	$(1-x)^{-n}$
(8) $(-1)^k C(n+k-1, k)$	$(1+x)^{-n}$
(9) $C(n+k-1, k)a^k$	$(1-ax)^{-n}$
(10) $\frac{1}{k!}$	e^x
(11) $\frac{(-1)^{k+1}}{k}$	$\ln(1+x)$

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Counting Problems and Generating Functions (*continued*)

Example: Use generating functions to find the number of k -combinations of a set with n elements, i.e., $C(n, k)$.

Solution: Each of the n elements in the set contributes the term $(1 + x)$ to the generating function

$$f(x) = \sum_{k=0}^n a^k x^k.$$

Hence $f(x) = (1 + x)^n$ where $f(x)$ is the generating function for $\{a^k\}$, where a^k represents the number of k -combinations of a set with n elements.

By the binomial theorem, we have

$$f(x) = \sum_{k=0}^n \binom{n}{k} x^k,$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Hence,

$$C(n, k) = \frac{n!}{k!(n-k)!}.$$

Example] Use generating functions to find the number of r -combinations from a set with n elements with repetitions allowed.

Solution:

Since there are n elements in the set, each can be selected zero times, one times and so on. It follows that

$$G(x) = (1 + x + x^2 + x^3 + \cdots)^n = \left(\frac{1}{1-x}\right)^n = \frac{1}{(1-x)^n}$$

the number of r -combinations from a set with n elements with repetitions allowed, is the coefficient a_r of x^r in the expansion of $G(x)$. Since

$$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$$

Then the coefficient a_r equals $C(n+r-1, r)$

【Example】 Use generating functions to find the number of r -combinations from a set of n elements with repetitions allowed.

Solution:

Since there are n elements, one time each, one time each, and so on, the generating function is

$$G(x) = (1 + x + x^2 + \cdots)^n$$

the number of r -combinations with repetitions allowed, is the coefficient a_r of x^r in the expansion of $G(x)$. Since

$$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$$

Then the coefficient a_r equals $C(n+r-1, r)$

$$(1-x)^{-n} = (1+(-x))^{-n} = \sum_{r=0}^{\infty} \binom{-n}{r} (-x)^r.$$

$$\begin{aligned} \binom{-n}{r} (-1)^r &= (-1)^r C(n+r-1, r) \cdot (-1)^r \\ &= C(n+r-1, r). \end{aligned}$$

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5. Using Generating Functions to Solve Recurrence Relations

The Methods of Solving Recurrence Relations

- ☐ Iterative approach
- ☐ Use a systematic way to solve an important class of recurrence relations
- ☐ Generating functions

Method:

- (1) Use the recurrence relation to find the generating function of this sequence;
- (2) $G(x) \leftrightarrow a_n$

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【Example】 Use generating functions to solve the recurrence relation $a_n = 2a_{n-1} + 3a_{n-2} + 4^n + 6$ with initial conditions $a_0 = 20, a_1 = 60$.

Solution:

$$\begin{aligned}
 a_n &= 2a_{n-1} + 3a_{n-2} + 4^n + 6 \quad \times x^n \\
 a_n x^n &= 2a_{n-1} x^n + 3a_{n-2} x^n + 4^n x^n + 6x^n \\
 \sum_{n=2}^{\infty} a_n x^n &= 2 \sum_{n=2}^{\infty} a_{n-1} x^n + 3 \sum_{n=2}^{\infty} a_{n-2} x^n + \sum_{n=2}^{\infty} 4^n x^n + 6 \sum_{n=2}^{\infty} x^n \\
 &\quad \swarrow \quad \downarrow \quad \downarrow \quad \swarrow \quad \searrow \\
 G(x) - a_0 - a_1 x &\quad 2x \sum_{n=1}^{\infty} a_n x^n \quad 3x^2 \sum_{n=0}^{\infty} a_n x^n \quad \frac{1}{1-4x} - 1 - 4x \quad 6\left(\frac{1}{1-x} - 1 - x\right) \\
 &\quad \downarrow \quad \downarrow \\
 &\quad 2x(G(x) - a_0) \quad 3x^2 G(x)
 \end{aligned}$$

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$$(1 - 2x - 3x^2)G(x) = \frac{20 - 80x + 2x^2 + 40x^3}{(1 - 4x)(1 - x)}$$

$$G(x) = \frac{20 - 80x + 2x^2 + 40x^3}{(1 - 4x)(1 - x)(1 + x)(1 - 3x)}$$

$$= \frac{16/5}{1-4x} + \frac{-3/2}{1-x} + \frac{31/20}{1+x} + \frac{67/4}{1-3x}$$

$$\frac{16}{5} \times 4^n - \frac{2}{3} \times 1^n + \frac{31}{20} \times (-1)^n + \frac{67}{4} \times 3^n$$

$$a_n = \frac{16}{5} \times 4^n - \frac{2}{3} + \frac{31}{20} \times (-1)^n + \frac{67}{4} \times 3^n$$

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6. Proving Identities via Generating Functions

The method of proving combinatorial identities:

- ❑ Use combinatorial proofs
- ❑ Use generating functions

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【Example 】 Use generating functions to prove Pascal's identity $C(n, r) = C(n-1, r) + C(n-1, r-1)$ when n and r are positive integers with $r < n$.

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Proof:

$$G(x) = (1+x)^n = \sum_{r=0}^n C(n,r)x^r$$

$$(1+x)^n = (1+x)(1+x)^{n-1} = (1+x)^{n-1} + x(1+x)^{n-1}$$

$$\sum_{r=0}^n C(n,r)x^r = \sum_{r=0}^{n-1} C(n-1,r)x^r + \sum_{r=0}^{n-1} C(n-1,r)x^{r+1}$$

$$= \sum_{r=0}^{n-1} C(n-1,r)x^r + \sum_{r=1}^n C(n-1,r-1)x^r$$

$$1 + \sum_{r=1}^{n-1} C(n,r)x^r + x^n = 1 + \sum_{r=1}^{n-1} C(n-1,r)x^r + \sum_{r=1}^{n-1} C(n-1,r-1)x^r + x^n$$

$$\sum_{r=1}^{n-1} \underline{C(n,r)}x^r = \sum_{r=1}^{n-1} [\underline{C(n-1,r) + C(n-1,r-1)}]x^r$$

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Homework

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